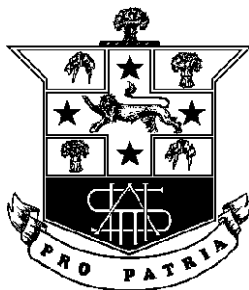


HURLSTONE AGRICULTURAL HIGH SCHOOL



YEAR 12

MATHEMATICS

EXTENSION 2

2010

HSC COURSE

HALF YEARLY EXAMINATION
(ASSESSMENT TASK 2)

EXAMINERS ~ G. RAWSON AND S. GEE

GENERAL INSTRUCTIONS

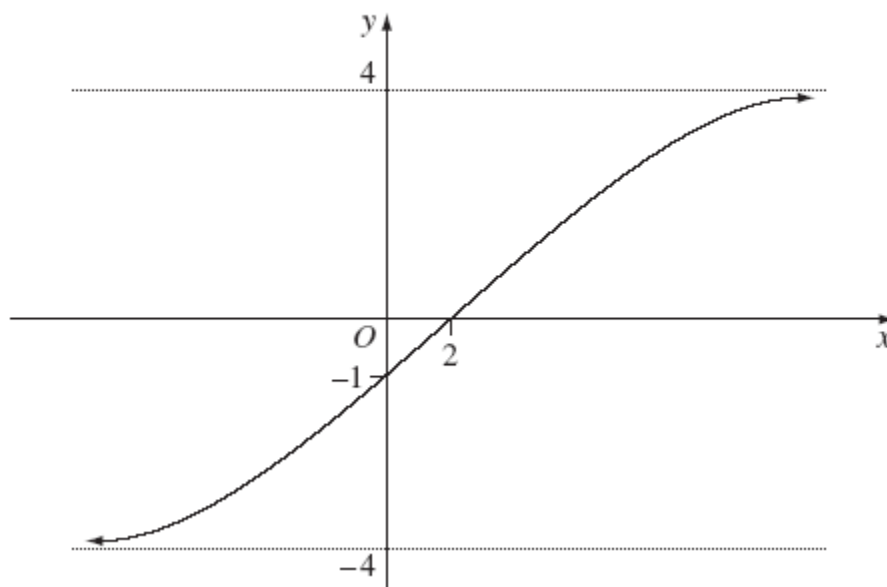
- | | |
|--|---|
| <ul style="list-style-type: none">• Reading Time – 5 minutes.• Working Time – Two Hours.• Attempt all questions.• Questions are of equal value.• All necessary working should be shown in every question.• This paper contains Five (5) questions.• Total Marks – 75 marks | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved Calculators and Templates may be used.• Each question is to be started in a new Examination Booklet.• This assessment task must NOT be removed from the Examination Room. |
|--|---|

STUDENT NAME / NUMBER: _____

TEACHER: _____

QUESTION ONE **15 marks** *Start a SEPARATE sheet*

- (a) The diagram shows the graph of $y = f(x)$.



Draw neat separate sketches of at least one third of a page for

- | | | |
|-------|-------------------|----------|
| (i) | $y = f(x+3)$ | 1 |
| (ii) | $y = f(x) $ | 1 |
| (iii) | $y = \sqrt{f(x)}$ | 1 |
| (iv) | $y = f(x)$ | 1 |
| (v) | $ y = f(x)$ | 1 |

- (b) Given the function $g(x) = (x-1)^2(x+2)^2$

Draw neat separate sketches of at least one third of a page for

- | | | |
|-------|----------------------|----------|
| (i) | $y = g(x)$ | 1 |
| (ii) | $y = \frac{1}{g(x)}$ | 1 |
| (iii) | $y = -g(x)$ | 1 |
| (iv) | $y = g(x) + 2$ | 1 |

Question 1 continued on next page...

Question 1 continued...

- (c) Given the function $f(x) = \frac{x^2 + 1}{x}$
- (i) Show $y = f(x)$ is an odd function. **1**
- (ii) Find the equations of any asymptotes. **1**
- (iii) Show $y = f(x)$ has a relative minimum point at $(1, 2)$. **1**
- (iv) Without using calculus explain why $y = f(x)$ has a relative maximum point at $(-1, -2)$. **1**
- (v) Sketch the curve $f(x) = \frac{x^2 + 1}{x}$ **1**
- (vi) With reference to your sketch in part (v),
explain why the inequation $\left| \frac{x^2 + 1}{x} \right| \leq 2$ has only two solutions. **1**

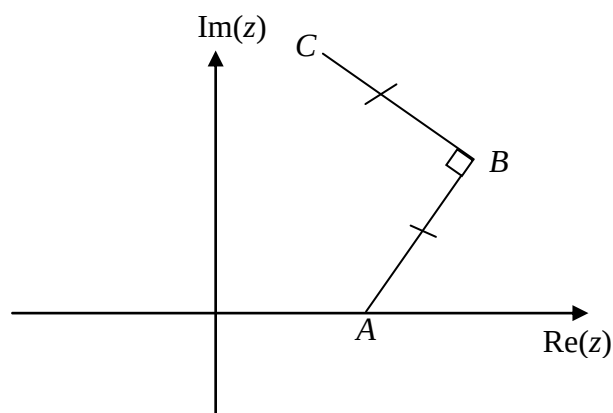
QUESTION TWO **15 marks** *Start a SEPARATE sheet*

(a) If $z = 3 - i$ and $w = 1 + 3i$, find (in the form $x + iy$) the simplest answer for

- | | | |
|-------|---------------|----------|
| (i) | $z - w$ | 1 |
| (ii) | zw | 1 |
| (iii) | $w\bar{w}$ | 2 |
| (iv) | $\frac{z}{w}$ | 2 |

- | | | | |
|-----|------|--|----------|
| (b) | (i) | Find the modulus and argument (in radians) of $5 + 5\sqrt{3}i$ | 2 |
| | (ii) | Hence, or otherwise, find the two square roots of $5 + 5\sqrt{3}i$ | 3 |

(c)

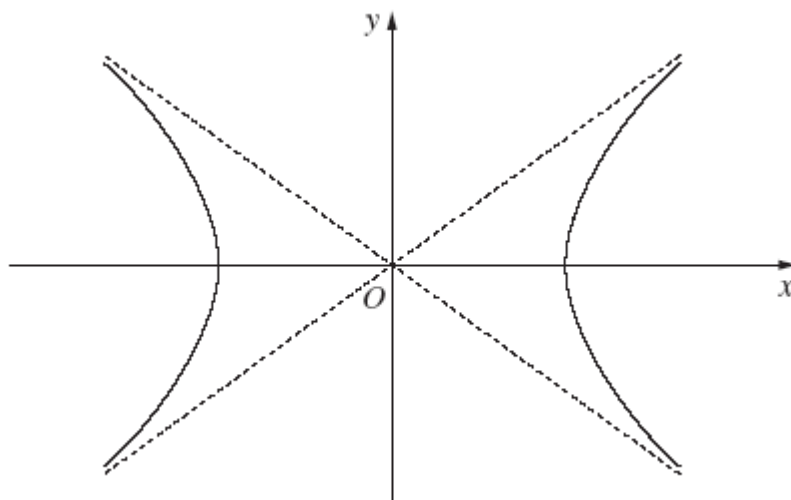


The diagram above shows the fixed points A , B and C in the Argand plane, where $AB = BC$, $\angle ABC = \frac{\pi}{2}$, and A , B and C are in anticlockwise order. The point A represents the complex number $z_1 = 2$ and the point B represents the complex number $z_2 = 3 + \sqrt{5}i$.

- | | | |
|------|---|----------|
| (i) | Find the complex number z_3 represented by the point C . | 2 |
| (ii) | D is the point on the Argand plane such that $ABCD$ is a square. Find the complex number z_4 represented by D . | 2 |

QUESTION THREE **15 marks** *Start a SEPARATE sheet*

- (a) The sketch shows a hyperbola in the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



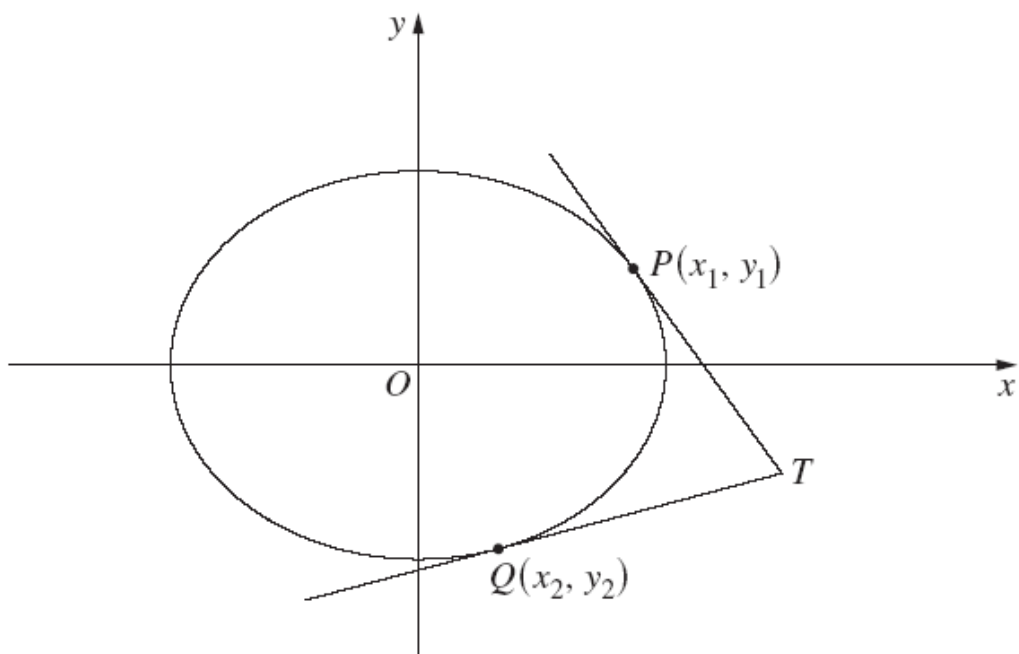
Given $a^2 = 144$ and $b^2 = 25$

- | | | |
|-------|---|----------|
| (i) | Find the coordinates of the points where the hyperbola meets the x - axis. | 1 |
| (ii) | Find the coordinates of the foci of the hyperbola. | 2 |
| (iii) | Find the equations of the asymptotes and directrices of the hyperbola. | 2 |
| (iv) | If a remains constant, what value must b become for the hyperbola to become rectangular? | 1 |
| (v) | Find the equation of the tangent to the curve at the point $P\left(13, \frac{1}{12}\right)$. | 1 |
| (vi) | Find the equation of the focal chord SP where S is the closest focus. | 1 |

Question 3 continued on next page...

Question 3 continued...

- (b) The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
The tangents at P and Q meet at T .



- (i) Show that the equation of the tangent at P is $\frac{x_1}{a^2}x + \frac{y_1}{b^2}y = 1$. 2
- (ii) Show that T lies on the line $\frac{(x_1 - x_2)}{a^2}x + \frac{(y_1 - y_2)}{b^2}y = 0$. 2
- (iii) Let M be the midpoint of PQ .
Show that O , M and T are collinear. 3

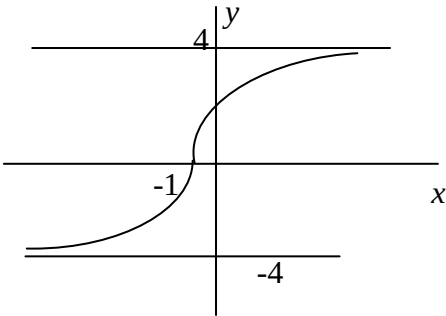
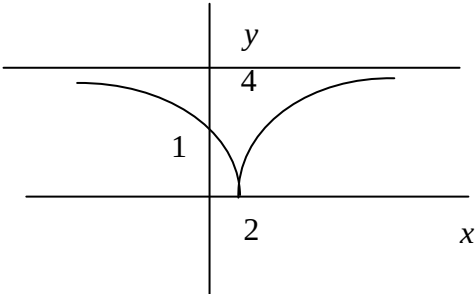
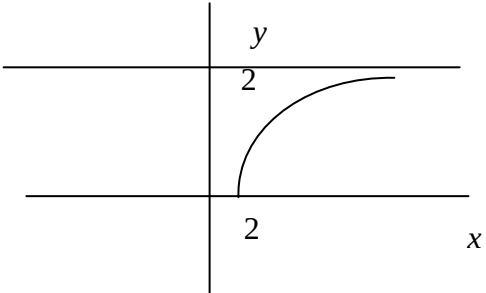
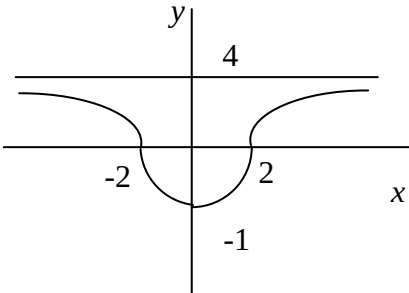
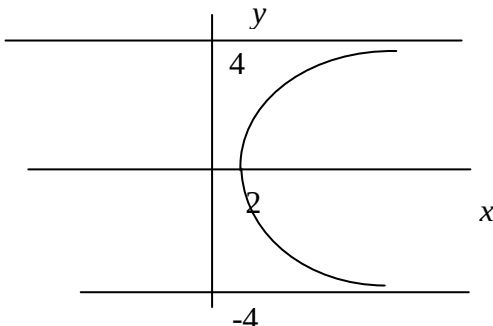
QUESTION FOUR **15 marks** *Start a SEPARATE sheet*

- (a) If $|z_1 - z_2| = |z_1 + z_2|$ show that $\arg z_1 - \arg z_2 = \frac{\pi}{2}$. 2
- (b) Plot on the Complex Plane, the values of z for which $z^3 - 8i = 0$. 3
- (c) What is the maximum value of $|z|$ for $|z - 1 - i| \leq 2$? 2
- (c) Sketch the locus of z satisfying:
- (i) $\arg(z - 4) = \frac{3\pi}{4}$ 3
- (ii) $\operatorname{Im} z = |z|$ 2
- (d) Given $z = x + iy$, sketch the locus of z if $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$, showing all important features. 3
- Write the Cartesian equation of this locus. 3

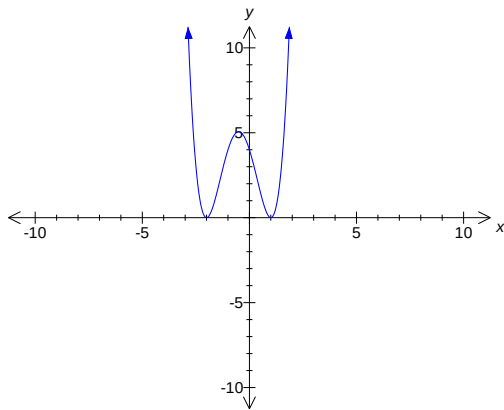
QUESTION FIVE 15 marks Start a SEPARATE sheet

- (a) Factorise $2^{n+1} + 2^n$ and hence write $\frac{2^{1001} + 2^{1000}}{3}$ as a power of 2. 2
- (b) Consider the circle $x^2 + y^2 - 2x - 14y + 25 = 0$
- (i) Show that if the line $y = mx$ intersects the circle in two distinct points, then $(1 + 7m)^2 - 25(1 + m^2) > 0$. 3
- (ii) For what values of m is the line $y = mx$ tangent to the circle? 2
- (c) Find the equation of $P(x)$ given it is monic of degree 5, has a double root at $x = 2$ and is odd. Sketch $y = P(x)$. 2
- (d) (i) Prove De Moivre's Theorem using mathematical induction.
ie. $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$, for integral values of $n \geq 0$. 3
- (ii) Show De Moivre's Theorem also holds for integral negative values. 2
- (iii) If $z = 1 + i\sqrt{3}$ find the principal argument of z^5 in radians. 1

END OF TEST

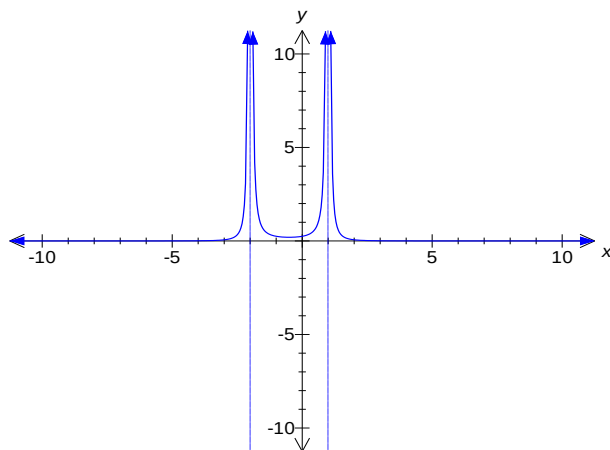
Year 12	Extension 2 Mathematics	HSC Task 2 Half Yearly 2010
Question No.1	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions		
Outcome	Solutions	Marking Guidelines
	(a)(i) $y = f(x+3)$ 	1 mark correct graph
	(ii) $y = f(x)$ 	1 mark correct graph
	(iii) $y = \sqrt{f(x)}$ 	1 mark correct graph
	(iv) $y = f(x)$ 	1 mark correct graph
	(v) $y = f(x)$ 	1 mark correct graph

b(i) $g(x) = (x-1)^2(x+2)^2$



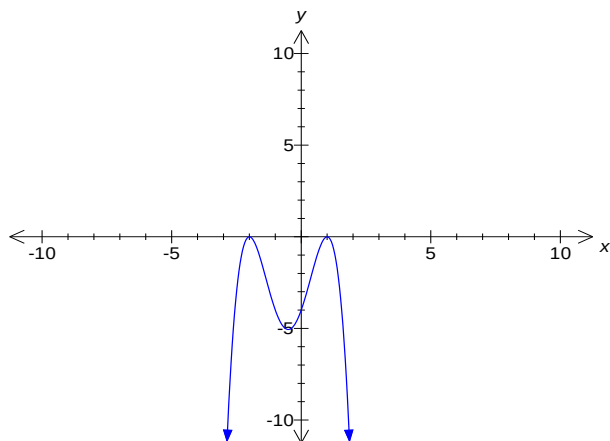
1 mark correct graph

(ii) $y = \frac{1}{g(x)}$



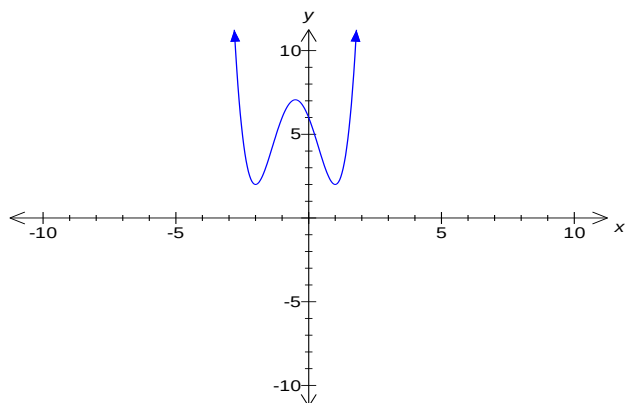
1 mark correct graph

(iii) $y = -g(x)$



1 mark correct graph

(iv) $y = g(x) + 2$



1 mark correct graph

C

(i)

Odd Function if $f(-x) = -f(x)$

$$f(x) = \frac{x^2 + 1}{x}$$

$$f(-x) = \frac{(-x)^2 + 1}{-x}$$

$$= -\frac{x^2 + 1}{x}$$

$$= -f(x) \therefore \text{odd function}$$

(ii)

Asymptotes $x = 0, y = x$

(iii)

$$f(x) = x + \frac{1}{x}$$

$$f(x) = x + x^{-1}$$

$$f'(x) = 1 - x^{-2}$$

Stationary points at $f'(x) = 0$

$$1 - \frac{1}{x^2} = 0$$

$$x^2 = 1$$

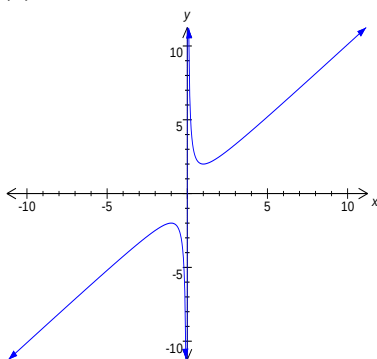
$$x = \pm 1$$

When $x = 1$ the point required is $(1, 2)$

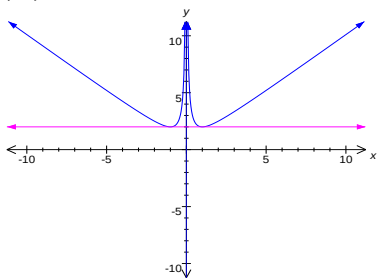
(iv)

Since is an odd function it has point symmetry through the origin. Therefore a relative minima will have a relative maxima as an image point and $(-1, -2)$ will be this point.

(v)



(vi)



Since the line $y=2$ cuts the curve above exactly twice and $y < 2$ has no intersecting points then there is exactly 2 solutions to the inequation

$$\left| \frac{x^2 + 1}{x} \right| \leq 2$$

1 mark correct answer

1 mark correct answer

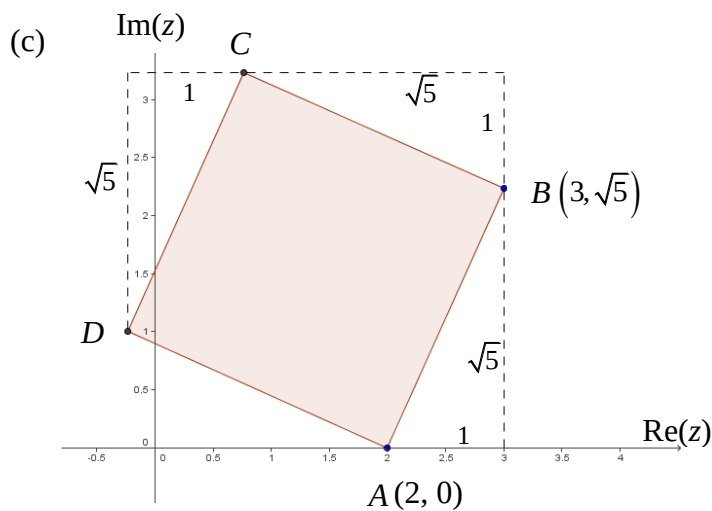
1 mark correct answer

1 mark correct reasoning

1 mark correct graph

1 mark correct reasoning

Year 12	Mathematics Extension 2	Half Yearly 2010 HSC
Question No. 2	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3 uses the relationship between algebraic and geometric representations of complex numbers		
Outcome	Solutions	Marking Guidelines
	(a) $z = 2 - i, w = 1 + 2i$ (i) $z - w = 3 - i - (1 + 3i)$ $= 2 - 4i$ (ii) $zw = (3 - i)(1 + 3i)$ $= 3 + 9i - i - 3i^2$ $= 6 + 8i$ (iii) $w\bar{w} = (1 + 3i)(\overline{1 + 3i})$ $= (1 + 3i)(1 - 3i)$ $= 1 - 9i^2$ $= 10$ (iv) $\frac{z}{w} = \frac{3 - i}{1 + 3i} \times \frac{1 - 3i}{1 - 3i}$ $= \frac{3 - 9i - i + 3i^2}{10}$ $= \frac{-10i}{10} = -i$ (b) (i) $z = \sqrt{5^2 + (5\sqrt{3})^2}$ $\arg(z) = \tan^{-1} \frac{5\sqrt{3}}{5}$ $= 10$ $= \frac{\pi}{3}$ (b) (ii) $(rcis\theta)^2 = 10cis\frac{\pi}{3}$ $r^2cis2\theta = 10cis\frac{\pi}{3}$ $r^2 = 10$ $2\theta = \frac{\pi}{3} \pm 2k\pi$ $r = \sqrt{10}$ $\theta = \frac{\pi}{6} \pm k\pi$ so the two square roots are $\sqrt{10}cis\frac{\pi}{6}$ and $\sqrt{10}cis\frac{-5\pi}{6}$ $\left\{ \text{or } \pm \frac{\sqrt{10}}{2}(\sqrt{3} + i) \right\}$ $\left\{ \begin{array}{l} (x + iy)^2 = 5 + 5\sqrt{3}i \\ x^2 - y^2 + 2xyi = 5 + 5\sqrt{3}i \\ \text{ie } x^2 - y^2 = 5 \\ \text{and } 2xy = 5\sqrt{3} \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} x = \frac{\sqrt{30}}{2}, -\frac{\sqrt{30}}{2} \\ y = \frac{\sqrt{10}}{2}, -\frac{\sqrt{10}}{2} \end{array} \right\}$	<u>1 mark</u> : correct answer <u>1 mark</u> : correct answer <u>2 marks</u> : correct answer <u>1 mark</u> : partially correct <u>2 marks</u> : correct answer <u>1 mark</u> : partially correct <u>2 marks</u> : correct solution <u>1 mark</u> : partially correct solution <u>3 marks</u> : correct solution <u>2 marks</u> : substantially correct solution <u>1 mark</u> : partially correct solution



- (i) $A \rightarrow B$ is across (right) 1, up $\sqrt{5}$
 $B \rightarrow C$ is up 1, across (left) $\sqrt{5}$
 So B is the point $(3 - \sqrt{5}, 1 + \sqrt{5})$
 i.e. the number $(3 - \sqrt{5}) + (1 + \sqrt{5})i$

2 marks : correct solution

1 mark : partially correct

- (ii) $C \rightarrow D$ is across (left) 1, down $\sqrt{5}$
 so D is the point $(2 - \sqrt{5}, 1)$
 i.e. the number $2 - \sqrt{5} + i$

2 marks : correct solution

1 mark : partially correct

Alternatively...

(i)
$$\begin{aligned}\overline{OC} &= \overline{OA} + \overline{AB} + \overline{BC} \\ &= 2 + (1 + \sqrt{5}i) + i(1 + \sqrt{5}i) \\ \therefore z_3 &= 3 - \sqrt{5} + (1 + \sqrt{5})i\end{aligned}$$

(ii)
$$\begin{aligned}\overline{OD} &= \overline{OA} + \overline{AD} \\ &= \overline{OA} + \overline{BC} \\ &= 2 + i(1 + \sqrt{5}i) \\ \therefore z_4 &= 2 - \sqrt{5} + i\end{aligned}$$

Year 12	Extension 2 Mathematics	HSC Task 2 Half Yearly 2010
Question No.3	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections.		
Outcome	Solutions	Marking Guidelines
	a $a^2 = 144, b^2 = 25$ $\frac{x^2}{144} - \frac{y^2}{25} = 1$ (i) $y = 0: \frac{x^2}{144} = 1$ $x^2 = 144$ $x = \pm 12$ (ii) $b^2 = a^2(e^2 - 1), \text{focii}(\pm ae, 0)$ $\frac{25}{144} = e^2 - 1$ $e^2 = \frac{169}{144}$ $e = \frac{13}{12}, e > 0$ $\text{focii at } \left(\frac{13}{12} \times 12, 0\right) \text{ and } \left(-\frac{13}{12} \times 12, 0\right)$ $(13, 0), (-13, 0)$ (iii) $\text{asymptotes: } y = \pm \frac{b}{a}x \quad y = \pm \frac{5}{12}x$ $\text{directrices: } x = \pm \frac{a}{e} \quad x = \pm \frac{12}{\frac{13}{12}} \quad \therefore x = \pm \frac{144}{13}$ (iv) $b = \pm 12, a = b \text{ for rectangular hyperbola}$ (v) $\text{tangent of form } \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ $\frac{13x}{144} - \frac{\frac{1}{12}y}{25} = 1$ $\frac{13x}{144} - \frac{y}{300} = 1$ (vi) $\text{Chord } SP \quad (13, 0), \left(13, \frac{1}{12}\right)$ $x = 13$	1 mark correct answer 2 marks correct method leading to correct conclusion 1 marks substantial progress in proof with correct method leading to an appropriate conclusion 1 mark correct answer asymptote 1 mark correct answer directrices 1 mark correct answer 1 mark correct answer 1 mark correct answer

b

(i)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

gradient of tangent at (x_1, y_1) : $m = -\frac{b^2 x_1}{a^2 y_1}$

equation of tangent at (x_1, y_1) : $y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$

$$a^2 y_1 y - a^2 y_1^2 = -b^2 x_1 x + b^2 x_1^2$$

$$a^2 y_1 y + b^2 x_1 x = b^2 x_1^2 + a^2 y_1^2$$

$$\frac{y_1 y}{b^2} + \frac{x_1 x}{a^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

$$\frac{y_1 y}{b^2} + \frac{x_1 x}{a^2} = 1 \text{ as } \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

(ii)

Similarly Q lies on $\frac{y_2 y}{b^2} + \frac{x_2 x}{a^2} = 1$

T is the point of intersection $\therefore$ solve simultaneously

$$\therefore \frac{y_1 y}{b^2} + \frac{x_1 x}{a^2} = \frac{y_2 y}{b^2} + \frac{x_2 x}{a^2}$$

$$\frac{x_1 x}{a^2} - \frac{x_2 x}{a^2} + \frac{y_1 y}{b^2} - \frac{y_2 y}{b^2} = 0$$

$$\frac{x(x_1 - x_2)}{a^2} + \frac{y(y_1 - y_2)}{b^2} = 0$$

(iii)

If M lies on line in Part (ii) then it should satisfy the equation

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Consider $\frac{x(x_1 - x_2)}{a^2} + \frac{y(y_1 - y_2)}{b^2}$

$$\frac{x_1 + x_2}{2} \times \frac{(x_1 - x_2)}{a^2} + \frac{(y_1 - y_2)}{b^2} \times \frac{y_1 + y_2}{2}$$

$$\frac{1}{2} \left[\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - \left(\frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} \right) \right]$$

$$\frac{1}{2} [1 - 1] = 0 \text{ as both } \frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} = 1 \text{ and } \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

$\therefore M$ lie on the same line as T from part (ii)

Substitute $(0,0)$ into $\frac{x(x_1 - x_2)}{a^2} + \frac{y(y_1 - y_2)}{b^2} = 0$

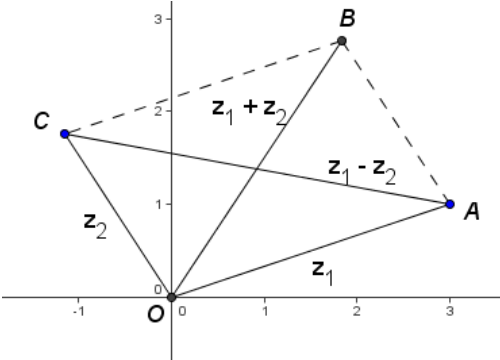
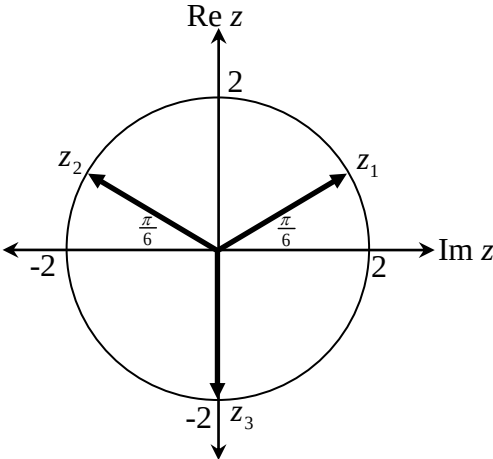
$$\frac{0 \times (x_1 - x_2)}{a^2} + \frac{0 \times (y_1 - y_2)}{b^2} = 0 \text{ which is true } \therefore O, M \text{ and } T \text{ are collinear.}$$

2 marks correct method leading to correct equation
1 marks substantial progress in proof with correct method leading to an appropriate conclusion

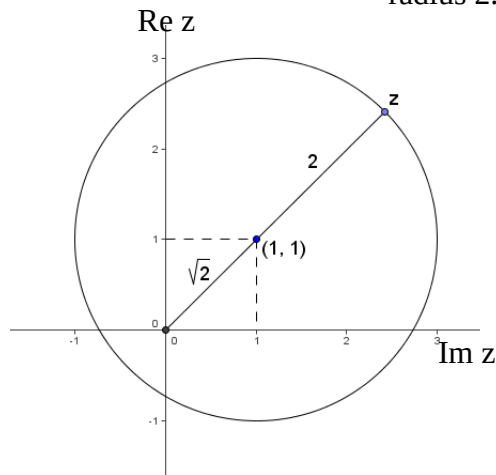
2 marks correct method with reasoning leading to correct conclusion
1 marks substantial progress in proof with correct method leading to an appropriate conclusion

3 marks correct method leading to correct conclusion
2 marks substantial progress in proof with correct method leading to an appropriate conclusion
1 mark elementary progress towards correct solution

Note if you used gradients you had to state they had a point in common therefore were not parallel but coincident

Year 12	Mathematics Extension 2	Half Yearly 2010 HSC
Question No. 4	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3	uses the relationship between algebraic and geometric representations of complex numbers	
Outcome	Solutions	Marking Guidelines
	(a)  $\overrightarrow{OB}$ represents $z_1 + z_2$ and $\overrightarrow{CA}$ represents $z_1 - z_2$. $z_1 - z_2 = z_1 + z_2 \Rightarrow OB = CA$ $\therefore OACB$ is a rectangle $\therefore z_1 \perp z_2$ so $\arg z_1 - \arg z_2 = \frac{\pi}{2}$ (b) $z^3 = 8i$ $r^3 \operatorname{cis} 3\theta = 8 \operatorname{cis} \frac{\pi}{2}$ so $r^3 = 8$ and $3\theta = \frac{\pi}{2} + 2k\pi$ $r = 2$ and $\theta = \frac{\pi}{6} + \frac{2k\pi}{3} \quad k = 0, \pm 1$ $k = 0 \Rightarrow z_1 = 2 \operatorname{cis} \frac{\pi}{6} = \sqrt{3} + i$ $k = 1 \Rightarrow z_2 = 2 \operatorname{cis} \frac{5\pi}{6} = -\sqrt{3} + i$ $k = -1 \Rightarrow z_3 = 2 \operatorname{cis} \frac{-\pi}{2} = -2i$ 	<u>2 marks</u> : correct solution <u>1 mark</u> : partially correct solution <u>3 marks</u> : correct solution <u>2 marks</u> : substantially correct solution <u>1 mark</u> : partially correct solution

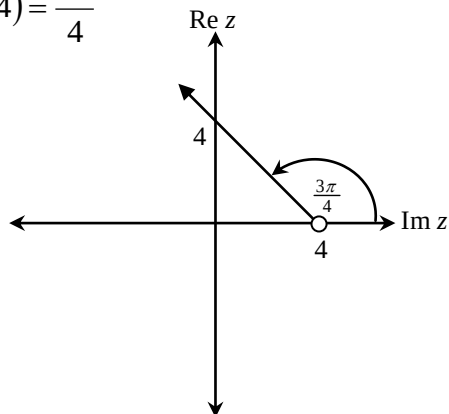
(c) $|z - 1 - i| \leq 2$ represents a circle, centre $(1, 1)$, radius 2.



Maximum value of $|z|$ is $2 + \sqrt{2}$

(d) (the second (c)... oops)

(i) $\arg(z - 4) = \frac{3\pi}{4}$



(ii) $\text{Im } z = |z|$

let $z = x + iy$

$\text{Im}(x + iy) = |x + iy|$

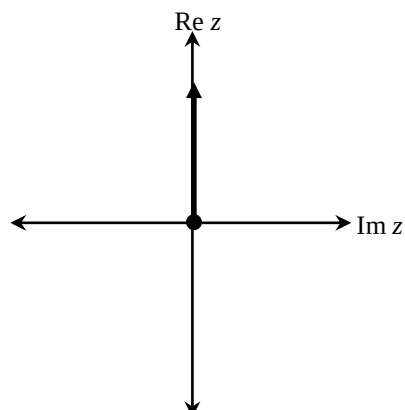
$y = \sqrt{x^2 + y^2}$

$y^2 = x^2 + y^2$

$x^2 = 0$

$x = 0$

note: as $|z| \geq 0$, $y \geq 0$



2 marks : correct solution

1 mark : progress towards correct solution

3 marks : correct solution

2 marks : substantially correct

1 mark : partially correct

2 marks : correct solution

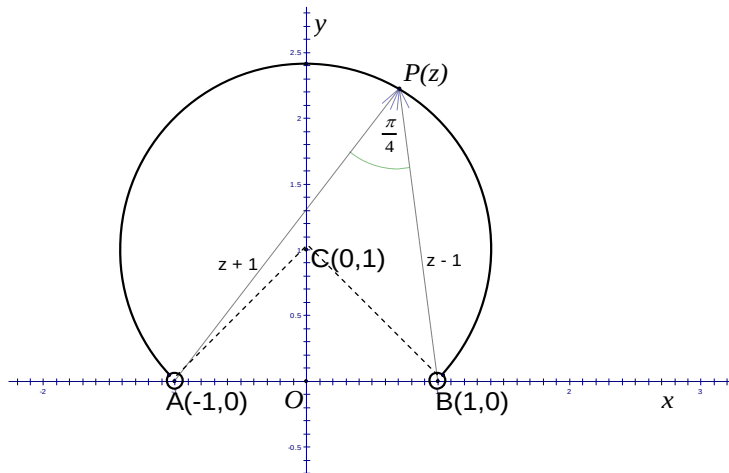
1 mark : progress towards correct solution

(e) ((d) on the paper)

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

$$\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$$

$$\arg(z+1) = \arg(z-1) - \frac{\pi}{4}$$



Cartesian equation is

$$x^2 + (y-1)^2 = 2, \text{ for } y > 0$$

OR, algebraically:

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1}$$

$$= \frac{[(x-1)+iy]}{[(x+1)+iy]} \cdot \frac{[(x+1)-iy]}{[(x+1)-iy]}$$

$$= \frac{[(x^2-1^2)+y^2] + [y(x+1)-y(x-1)]i}{(x+1)^2 + y^2}$$

$$= \frac{[x^2+y^2-1]}{(x+1)^2 + y^2} - \frac{2y}{(x+1)^2 + y^2}i$$

$$\tan\left[\arg\left(\frac{z-i}{z+i}\right)\right] = \frac{-\frac{2y}{(x+1)^2 + y^2}}{\frac{x^2+y^2-1}{(x+1)^2 + y^2}}$$

$$= -\frac{2y}{x^2+y^2-1}$$

$$\text{as } \tan \frac{\pi}{4} = 1$$

$$-\frac{2y}{x^2+y^2-1} = 1$$

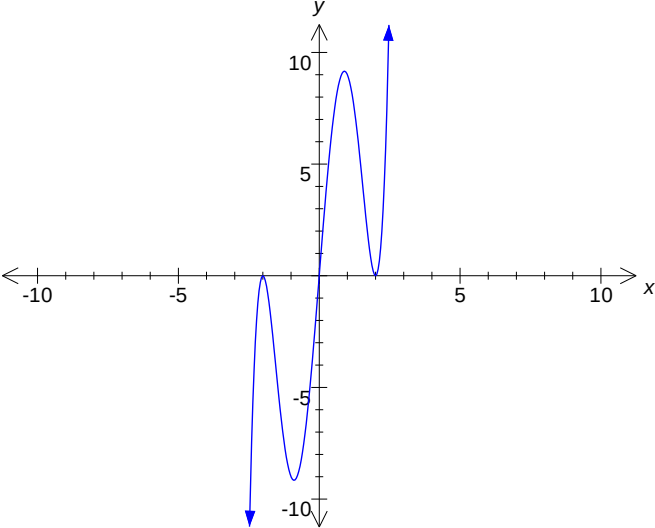
so the relation is $x^2 + y^2 + 2y - 1 = 0$

Note: this method alone will not show that the locus is above the x-axis only.

3 marks : correct solution

2 marks : substantially correct

1 mark : partially correct

Year 12	Extension 2 Mathematics	HSC Task 2 Half Yearly 2010
Question No.5	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings		
Outcome	Solutions	Marking Guidelines
	a $2^{n+1} + 2^n = 2 \times 2^n + 2^n$ $= 2^n (2 + 1)$ $= 3 \times 2^n$ $\frac{2^{1001} + 2^{1000}}{3} = \frac{3(2^{1000})}{3}$ $= 2^{1000}$ b (i) $x^2 + y^2 - 2x - 14y + 25 = 0$ $y = mx$ Solve simultaneously and consider discriminants $x^2 + (mx)^2 - 2x - 14(mx) + 25 = 0$ $(m^2 + 1)x^2 - 2(1 + 7m)x + 25 = 0$ for two solutions $\Delta > 0$ $4(1 + 7m)^2 - 4 \times (m^2 + 1) \times 25 > 0$ $(1 + 7m)^2 - (m^2 + 1) \times 25 > 0$ (ii) for tangent $\Delta = 0$ $(1 + 7m)^2 - (m^2 + 1) \times 25 = 0$ $49m^2 + 14m + 1 - 25m^2 - 25 = 0$ $24m^2 + 14m - 24 = 0$ $12m^2 + 7m - 12 = 0$ $(3m + 4)(4m - 3) = 0$ $m = \frac{-4}{3}, \frac{3}{4}$ c $P(x) = x(x - 2)^2(x + 2)^2$ 	1 mark correct index expression 1 mark correct answer 3 marks correct method leading to correct conclusion 2 marks substantial progress in proof with correct method leading to an appropriate conclusion 1 mark elementary progress towards correct solution 2 marks correct method leading to correct conclusion 1 marks substantial progress in proof with correct method leading to an appropriate conclusion 1 mark correct answer 1 mark correct graph

d (i)

Use Induction to prove $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

Prove True for $n = 0, 1$

$$LHS : (\cos \theta + i \sin \theta)^0 = 1$$

$$RHS : \cos 0 \times \theta + i \sin 0 \times \theta = 1$$

$$LHS = RHS$$

$$\text{also } n = 1 : \cos \theta + i \sin \theta = \cos \theta + i \sin \theta$$

$\therefore$ true for $n = 0, 1$

Assume true for $n = k$

$$\text{i.e. } (\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

Prove true for $n = k + 1$

$$\text{i.e. } (\cos \theta + i \sin \theta)^{k+1} = \cos(k+1)\theta + i \sin(k+1)\theta$$

$$LHS : (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta)$$

$$(\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta)$$

equating the real and the imaginary parts

$$\cos k\theta \cos \theta - \sin k\theta \sin \theta + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta)$$

$$\cos(k+1)\theta + i \sin(k+1)\theta$$

$$= RHS$$

$\therefore$ if true for $n = k$ is true for $n = k + 1$

Since proved true for $n = 1$ and $n = k + 1$ is true for $n = 2, 3, \dots$

and hence by the process of mathematical induction

is true for all positive integral values of n .

ii

$$(\cos \theta + i \sin \theta)^{-n} = \frac{1}{(\cos \theta + i \sin \theta)^n}$$

$$\text{by part (i)} \quad = \frac{1}{(\cos n\theta + i \sin n\theta)}$$

realising the denominator

$$= \frac{1}{(\cos n\theta + i \sin n\theta)} \times \frac{(\cos n\theta - i \sin n\theta)}{(\cos n\theta - i \sin n\theta)}$$

$$= \frac{(\cos n\theta - i \sin n\theta)}{\cos^2 n\theta + \sin^2 n\theta}$$

$$= \cos n\theta - i \sin n\theta$$

$$= \cos(-n\theta) + i \sin(-n\theta)$$

$\therefore$ true for negatives

iii

$$\arg(z) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

$$\arg(z^5) = 5 \times \frac{\pi}{3} = -\frac{\pi}{3} \text{ (principle argument)}$$

3 marks correct method
leading to correct conclusion
2 marks substantial progress in
proof with correct method
leading to an appropriate
conclusion
1 mark elementary progress
towards correct solution

3 marks correct method
leading to correct conclusion
2 marks substantial progress in
proof with correct method
leading to an appropriate
conclusion
1 mark elementary progress
towards correct solution

1 mark correct solution